Data Structures and Algorithms -3 (25CS2103E)

Automated Project Task Planning and Scheduling

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Abstract

- Develops a **Project Scheduling System** to manage tasks and their dependencies.
- Represents tasks and dependencies using a **Directed Graph (DAG)**.
- Uses **Topological Sorting** to generate a valid task execution order.
- Calculates task completion times while satisfying **precedence constraints**.
- Identifies the **Critical Path** and critical tasks affecting project completion.
- Helps determine the **minimum overall project completion time**.
- Demonstrates the application of **graph algorithms and optimization techniques** in real-world project management.
- Designed to efficiently handle **up to 1000 tasks**.

Introduction

- Project management involves handling multiple tasks that may have **dependencies and execution constraints**.
- A task can begin only after its required predecessor tasks are completed.
- Managing these dependencies manually becomes difficult as the **number of tasks increases**.
- The proposed system represents tasks and dependencies using a **Directed Graph**.
- **Topological Sorting** is used to determine a valid execution sequence that satisfies all dependencies.
- Task durations are considered to calculate the **project completion time and critical path**.
- The system demonstrates the use of **graph algorithms and optimization techniques** for real-world scheduling problems.
- It supports efficient scheduling for projects containing **up to 1000 tasks**.

Literature Survey

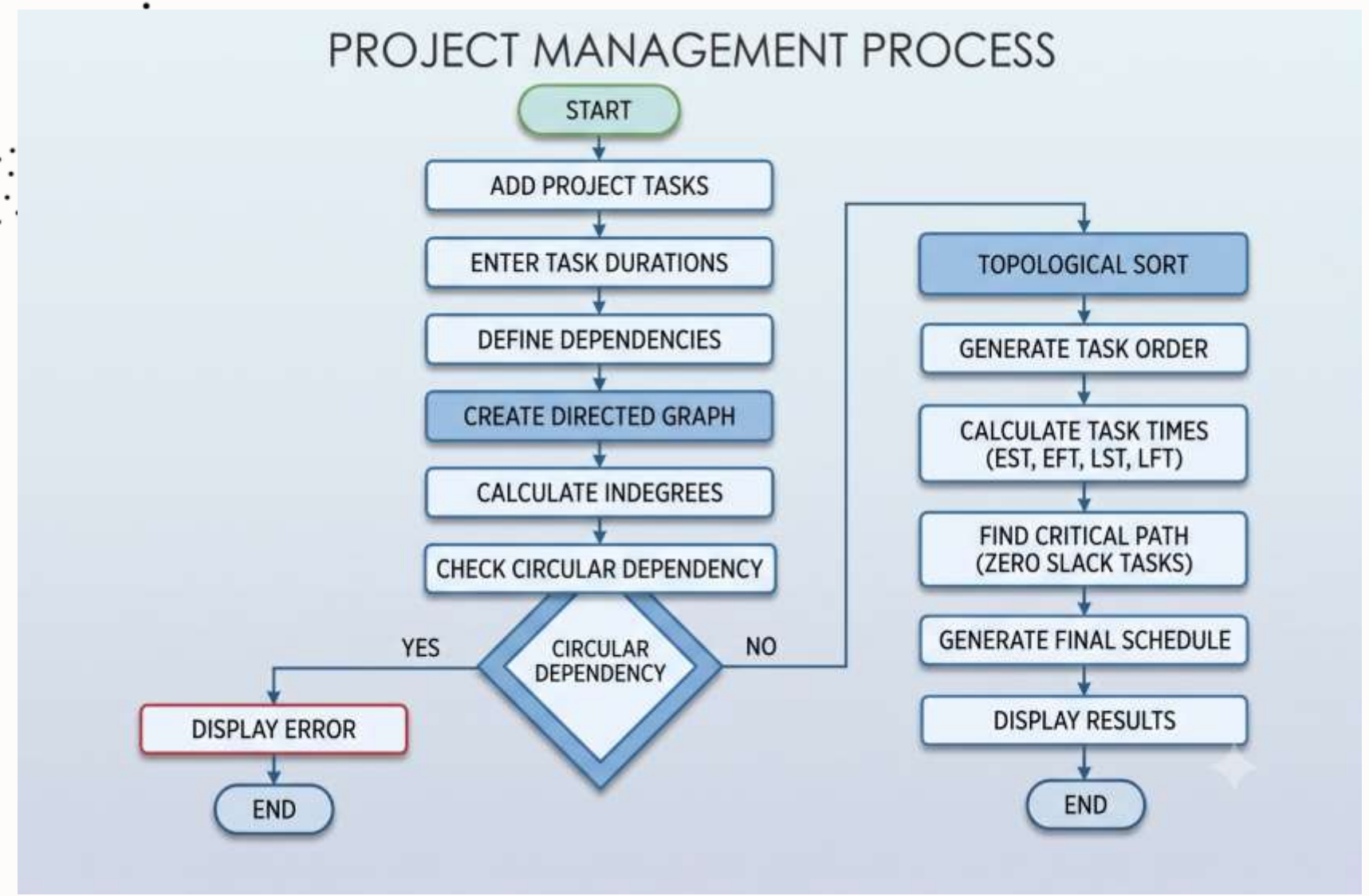
- **Graph-Based Scheduling:** Project tasks can be represented as vertices and dependencies as directed edges. Topological sorting helps generate a valid task execution sequence.
- **Critical Path & Optimization:** Scheduling approaches analyze task durations and dependencies to identify critical tasks and reduce overall project completion time.
- **Advanced Scheduling:** The DSA-3 course recognizes scheduling as an optimization problem and includes computational complexity and approximation techniques for scheduling.

Problem Statement and Objectives

- **Problem Statement:**
- Project tasks often have **dependencies and precedence constraints** that determine their execution order.
- Manually managing these dependencies becomes difficult as the **number of tasks increases**.
- The system aims to generate a **valid and optimized execution schedule** while satisfying all task dependencies.
- It also identifies **critical tasks and the overall project completion time**.

- **Objectives:**
- **Add and manage** project tasks, durations, and dependencies.
- Use **Graph and Topological Sorting** to determine the correct execution order.
- **Detect circular dependencies** that make scheduling impossible.
- Calculate the **minimum project completion time**.
- Identify the **Critical Path and critical tasks**.
- Efficiently handle projects with **up to 1000 tasks**.

Flow Chart


PROJECT MANAGEMENT PROCESS
START
ADD PROJECT TASKS
ENTER TASK DURATIONS
DEFINE DEPENDENCIES
CREATE DIRECTED GRAPH
CALCULATE INDEGREES
CHECK CIRCULAR DEPENDENCY
CIRCULAR DEPENDENCY
YES
NO
DISPLAY ERROR
END
TOPOLOGICAL SORT
GENERATE TASK ORDER
CALCULATE TASK TIMES (EST, EFT, LST, LFT)
FIND CRITICAL PATH (ZERO SLACK TASKS)
GENERATE FINAL SCHEDULE
DISPLAY RESULTS
END

Results

- Successfully generated a **valid project execution schedule** while satisfying all task dependencies.
- **Topological Sorting** determined the correct order of task execution and helped identify circular dependencies.
- Task durations were analyzed to calculate the **minimum project completion time**.
- The **Critical Path and critical tasks** were identified for effective project management.
- The system demonstrates the application of **graph algorithms and optimization techniques** to project scheduling.

Hardware and Software requirements

- **Hardware Requirements:**
- **Processor:** Intel Core i3 or equivalent
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB free space
- **Display:** Standard monitor
- **Keyboard & Mouse**

- **Software Requirements:**
- **Operating System:** Windows / Linux
- **Programming Language:** Java
- **IDE:** IntelliJ IDEA / Eclipse / VS Code
- **JDK:** JDK 17 or above
- **Data Structures:** Graph, Adjacency List, Queue
- **Algorithms:** Topological Sorting and Critical Path Analysis
- **No special hardware required**

Conclusion

- Successfully developed a **Project Scheduling System** using graph-based techniques.
- **Topological Sorting** generates a valid task execution order while satisfying all dependencies.
- The system identifies the **Critical Path** and calculates the minimum project completion time.
- The project demonstrates the practical use of **DSA and optimization techniques** in project management.

Future Scope

- Add **resource constraints and task priorities** for more realistic scheduling.
- Support **deadlines and multiple project resources** to improve optimization.
- Integrate **advanced approximation algorithms** for complex scheduling problems.
- Add **parallel task execution** to reduce overall project completion time.

(DEEMED TO BE UNIVERSITY)

Thank you